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Neural network project (Review Sentiment Classification)

Overview

In this project, we explored **four different approaches** for review sentiment classification:

1. Sentence Embedding + Dense Layer (**best performance**)
2. Word Embedding + Transformer (**overfitting**)
3. Word Embedding + 1D CNN
4. Word Embedding + GRU

Sentence Embedding + Dense Layer (Best Approach)

- We used the **`all-mpnet-base-v2`** sentence embedder to generate vector representations of reviews.
- The vectors were classified using a **dense layer ending with SoftMax**.
- Initial results were good but **overfitting** and **imbalanced data** reduced performance.

Solutions Applied:

1. **Sample weighting** to handle class imbalance.
2. **Data augmentation**:

    - **Synonym replacement**
    - **Resampling**: duplicate small classes, down sample large classes
3. Preprocessing:
    - Replace URLs with the token `"URL"`
    - Convert all text to lowercase
    - Remove extra spaces and punctuation

Conclusion:

- **Imbalanced data** was the main challenge.
- With proper augmentation and embedding, the model's accuracy improved significantly.

Word Embedding + Transformer (Overfitting)

- Built a **transformer encoder** model (like GPT encoder but only using the encoder).
- Input: embedded words.

Challenges:

- The model **overfits** easily.
- Adding **regularization** had minimal effect.
- Data augmentation (WordNet synonym replacement, noise, back-translation, paraphrasing) caused:
    - **Underfitting** for paraphrasing
    - Long processing time for back-translation

Observations:

- Transformers work better with **real-world text** without removing stopwords.
- They perform well with **large datasets** and minimal preprocessing.

Word Embedding + 1D CNN

- Built a 1D CNN for sequence classification.

Challenges:

- Model **overfitted** quickly.
- Tried smart augmentation:
    - Down sample major classes
    - Up sample minor classes
- Preprocessing pipeline:

- Remove stop words carefully (e.g., keep “not” by splitting contractions: `isn't → is not`)
- Apply lemmatization

Conclusion:

- 1D CNN was **too simple** to capture the full relationships between words in a sentence.

Word Embedding + GRU

- Built a GRU-based sequential model.

Challenges:

- Overfitting remains an issue.
- Dropout reduced overfitting but **slowed GPU training**, forcing CPU usage.

Observations:

- GRU performed **better than 1D CNN**, but complexity increased due to sequential processing.

Overall Conclusions

1. **Sentence embeddings + dense layers** achieved the **best results** after proper augmentation and handling class imbalance.
2. **Transformers** require **real-world data** and large datasets; they are sensitive to preprocessing.
3. **CNNs** are too simple for semantic relationships in sentences.
4. **GRUs** capture sequential information better but increase complexity and may slow GPU training.
5. Handling **imbalanced datasets** and applying **smart augmentation** are crucial to improve accuracy.